

for the field of wearables as it opens up a host of possibilities for easier, more intuitive UI.

Let's take a look at some of the innovations in UI and how they can be implemented towards wearables.

Gesture Interfaces

Gesture-based Interfaces rely on bodily motion for input, sometimes augmented by voice-based input. Probably one of the coolest looking ones, this type of interface was best conceptualized in the 2002 Sci-fi hit, *Minority Report*, in which Tom Cruise was shown to perform a number of complicated computing tasks such as manipulating images and videos using his hands, which are fitted with some futuristic gloves that seemingly act as sensors.

What seemed like far-fetched science fiction is now slowly becoming reality, thanks in part to the popularity of motion-sensing game controllers such as those seen in the Nintendo Wii and PlayStation Move as well as Microsoft's patented Kinect Motion Sensor, which also recognized voice-activated commands for a very seamless and intuitive user experience.

These devices use a collection of sensors to detect bodily motions when are then translated to input signals that drive the UI. The Wiimote and PlayStation Move controller use accelerometers and gyroscopes to translate the motions made by the controller while the Kinect uses a motion-sensing camera that is programmed to detect movements made by the user's limbs and head.

One can fathom the vast pool of possibilities and applications that Gesture-based interfaces hold in terms of wearables. Technology users have been for the longest time restricted to moving their point of interaction across a two-dimensional field. Think of running your mouse across your monitor. Flat, isn't it? Now imagine how useful the addition of a third dimension would be to what has traditionally been a flat interface. You could pull an object towards you to open their contents or fling them across the screen to archive or delete them. Bring two-handed gestures into the picture and you've added another gamut of interaction types. You can stretch an object to zoom in, clap across it to close, double tap on it to pause—we are only restricted by our imagination.

Currently, gesture-based interfaces are extremely popular for wearables that lack a screen. Also, wearables can double up as input devices in such gesture-based systems. Take, for example, Flying Fingers Mouse, a device that looks suspiciously like the glove worn by Tom Cruise in *Minority*